

# Shiyu Chen 陈时宇

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**Research interests:** autonomous exploration & active perception · semantic mapping & LiDAR–camera fusion · next-best-view planning · VLM-in-the-loop industrial inspection

## EDUCATION

**Carnegie Mellon University** — M.S. in Mechanical Engineering (Research) Aug 2025 – May 2027 (expected)  
*Computational Engineering and Robotics Lab (CERLAB), advised by Prof. Kenji Shimada* Pittsburgh, PA

- Focus: 5-DoF semantic exploration and active inspection for legged robots in industrial environments.

**Univ. of Cincinnati & Chongqing University** — Joint Program, B.S. / B.Eng. in Mechanical Engineering 2020 – 2025

*Robotics Minor (UC) · UC GPA 3.71/4.0 · CQU GPA 3.58/4.0 · Dean's List ×4* Cincinnati, OH / Chongqing, CN

## RESEARCH EXPERIENCE

**Graduate Researcher**, CERLAB Roboteam (CMU × YKK factory-automation inspection) Aug 2025 – Present

### Real-time LiDAR–camera semantic registration on a legged platform

- Built a real-time pipeline fusing Ouster LiDAR geometry with Insta360 X4 panoramic imagery on a Boston Dynamics Spot — dense semantically-colored point clouds at **10 FPS**, **<100 ms latency**, **<10 px reprojection error**, spatially consistent under gait vibration; designed a 3D-printed rigid sensor mount for stable extrinsic calibration.

**5-DoF semantic exploration with VLM-guided active inspection** (co-first author · arXiv:2609.19460 · submitted to ICRA 2027)

- Co-first-authored a legged-robot semantic exploration system in which Spot tilts its body — **5-DoF viewpoints (x, y, yaw + pitch, roll)** — to see the upper surfaces a level camera misses: postures sampled around each mapped object are chosen by expected coverage gain, executed aim-aligned, and merged with frontier viewpoints in one global planner (C++/ROS).
- Developed the **VLM-assisted viewpoint pruning** (observation history + bird's-eye-view map) that drops redundant inspection visits, with a geometric fallback so the autonomy stack never blocks on the model call.
- Built the Isaac Sim benchmark behind the paper (three industrial scenes, ground-truth surfaces, Dockerized): **+8–10 pp** coverage and **17–32% less exploration time** than the planar 3-DoF baseline, **53–73% fewer postures** than competing 5-DoF methods, highest mean object-coverage AUC; demonstrated on a real Spot in a machine shop (five machines).

**Undergraduate Researcher**, Cable-Driven Soft Robotic Arm for Minimally Invasive Surgery Sep 2024 – May 2025

*Univ. of Cincinnati, advised by Prof. Janet (Jiangxia) Dong*

- Designed and prototyped a wire-actuated continuum arm with **0.27 mm/step** resolution and **>230° bending range**; modeled nonlinear actuation and led hardware–control integration (motor selection, microstepping, Ecoflex validation) with clinical end-users; published in *ASME J. Medical Diagnostics* (2026).

**Research Assistant**, Deep-Learning Defect Detection in Composite Materials Jan 2022 – Jul 2022  
*Chongqing University, advised by Prof. Bo Yang*

- Developed CNN models (LeNet/VGG16) with multi-scale feature extraction and DSP-based analysis for defect detection/localization in carbon-fiber and welded parts, reaching **>97% accuracy**; applied CenterNet/EfficientDet on image datasets.

## PUBLICATIONS & PATENTS

X. Zhan\*, **S. Chen\***, K. Shimada, "Pose-aware Legged Robot Semantic Exploration with Omnidirectional Perception in Confined Unknown Environments," arXiv:2609.19460, Sep 2026. Submitted to ICRA 2027 (under review). \*Equal contribution.

Y. Wang, C. Liu, K. Jiang, B. Wu, **S. Chen**, J. Dong, A. Ashfaq, "Innovative Flexible Robotic Arm for Enhanced Precision in Transaortic Surgical Myectomy," *ASME Journal of Medical Diagnostics*, 9(2): 021001, 2026.

L. Liu, **S. Chen**, Z. Tang, "Intelligent Diagnosis of Bearing Faults Based on SDP Image Fusion," *Int. Conf. on Image Processing, Computer Vision and Machine Learning (ICICML)*, Chengdu, 2023.

Patents (China, as co-inventor): 4 granted invention patents (2025–2026) on deep-learning machine vision for steelmaking process monitoring (steel-stream landing-point, slag-splashing and slag-discharge detection); 3 pending applications in belt-conveyor inspection.

## INDUSTRY EXPERIENCE

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**Computer Vision R&D Assistant**, CISDI Information Technology  
Chongqing, China

May 2022 – Dec 2023

- Trained/optimized PyTorch detection models for an industrial belt-conveyor inspection system and deployed them to edge devices (NVIDIA Xavier NX / AGX Orin) via TensorRT, cutting inference latency by **40%**; ported Python algorithms to C++; adapted models to Huawei Ascend (2nd prize, Ascend AI competition); built a PyQt annotation tool and led labeling of 100k+ images. Received return offer.

## TECHNICAL SKILLS

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|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>     | C++ (proficient), Python (proficient), MATLAB, C                                                                                                           |
| <b>Robotics / Sim</b>  | ROS1 & ROS2, NVIDIA Isaac Sim, Spot SDK, LiDAR–camera calibration, frontier exploration, next-best-view / active perception, SLAM, path planning (TSP/LKH) |
| <b>Perception / ML</b> | PyTorch, TensorRT, YOLOv7, SAM2, VLM APIs, CNN/deep learning, point-cloud processing (ICP)                                                                 |
| <b>Tools</b>           | Linux, Docker, Git, SolidWorks, COMSOL                                                                                                                     |

## HONORS & TEACHING

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- National Bronze Prize**, 8th China Int'l "Internet+" Innovation & Entrepreneurship Competition, 2023.
- Teaching Assistant — Engineering Design Thinking II (UC, 2025) and C++ Programming (CQU, 2024).